

House File 2540 - Introduced

HOUSE FILE 2540
BY COMMITTEE ON EDUCATION

(SUCCESSOR TO HSB 610)

A BILL FOR

1 An Act relating to computer science and artificial intelligence
2 education, including high school curricula and graduation
3 requirements, and practitioner preparation programs, and
4 including applicability provisions.
5 BE IT ENACTED BY THE GENERAL ASSEMBLY OF THE STATE OF IOWA:

1 Section 1. Section 256.7, subsection 26, paragraph a,
2 unnumbered paragraph 1, Code 2026, is amended to read as
3 follows:

4 Adopt a core curriculum and adopt rules that establish high
5 school graduation requirements for all students in school
6 districts and accredited nonpublic schools that include at a
7 minimum satisfactory completion of four years of English and
8 language arts, three years of mathematics, three years of
9 science, ~~and three years of social studies, and, beginning with~~
10 students in the 2030-2031 school year graduating class, one
11 semester of computer science and artificial intelligence, or
12 the trimester or quarter equivalent, which may include computer
13 science instruction provided in grade eight pursuant to
14 section 256.11, subsection 4. One semester, or the trimester
15 or quarter equivalent, of computer science and artificial
16 intelligence instruction shall satisfy the graduation
17 requirement associated with both computer science and
18 artificial intelligence instruction and one semester, or the
19 trimester or quarter equivalent, of the graduation requirement
20 associated with either mathematics or science.

21 Sec. 2. Section 256.7, subsection 26, paragraph a,
22 subparagraph (4), Code 2026, is amended to read as follows:

23 (4) The rules shall provide for the establishment of
24 high-quality standards for computer science and artificial
25 intelligence education taught by elementary, middle, and high
26 schools, in accordance with the goal established under section
27 284.6A, subsection 1, setting a foundation for personal and
28 professional success in a high-technology, knowledge-based Iowa
29 economy. The high-quality standards for computer science and
30 artificial intelligence education must require that all levels
31 and grades of computer science and artificial intelligence
32 instruction include instruction related to the fundamental
33 concepts of evolving computer science technologies and
34 artificial intelligence, including instruction that explains
35 what artificial intelligence is, how artificial intelligence

1 works, and the societal and ethical impacts of artificial
2 intelligence.

3 Sec. 3. Section 256.9, subsection 61, Code 2026, is amended
4 to read as follows:

5 61. Develop and implement a statewide kindergarten through
6 grade twelve computer science and artificial intelligence
7 instruction plan ~~by July 1, 2022.~~

8 Sec. 4. Section 256.9, Code 2026, is amended by adding the
9 following new subsections:

10 NEW SUBSECTION. 80. On or before June 30, 2027, publish
11 on the department's internet site a list of computer science
12 and artificial intelligence courses that satisfy the rules
13 adopted by the state board pursuant to section 256.7,
14 subsection 32, paragraph "c", and that a school district
15 or accredited nonpublic school may use to implement the
16 graduation requirement described in section 256.11, subsection
17 5, paragraph "1", subparagraph (2). The list must include
18 the names of the computer science and artificial intelligence
19 courses and the course codes associated with each computer
20 science and artificial intelligence course.

21 NEW SUBSECTION. 81. Publish on the department's internet
22 site all of the information submitted to the department by
23 school districts pursuant to section 256.11, subsection 5,
24 paragraph "1", subparagraph (3).

25 NEW SUBSECTION. 82. On or before June 30, 2027, create
26 and publish on the department's internet site a plan to
27 expand statewide computer science and artificial intelligence
28 teacher capacity to support implementation of the graduation
29 requirement described in section 256.11, subsection 5,
30 paragraph "1", subparagraph (2). The plan must specifically
31 address targeted support for schools with a total enrollment of
32 less than five hundred students in grades nine through twelve.
33 When creating the plan, the department shall invite input from
34 representatives of kindergarten through grade twelve schools,
35 including students, computer science teacher professional

1 development providers, industries that employ technology
2 workers, and higher education institutions that offer computer
3 science teacher preparation pathways.

4 Sec. 5. Section 256.11, subsection 3, paragraph b, Code
5 2026, is amended to read as follows:

6 b. Computer science and artificial intelligence instruction
7 incorporating the standards established under section 256.7,
8 subsection 26, paragraph "a", subparagraph (4), shall be
9 offered in at least one grade level ~~commencing with the school~~
10 ~~year beginning July 1, 2023.~~

11 Sec. 6. Section 256.11, subsection 4, paragraph b, Code
12 2026, is amended to read as follows:

13 b. Computer science and artificial intelligence instruction
14 incorporating the standards established under section 256.7,
15 subsection 26, paragraph "a", subparagraph (4), shall be
16 offered in at least one grade level ~~commencing with the~~
17 ~~school year beginning July 1, 2023. Career exploration~~
18 ~~and development shall be designed so that students are~~
19 ~~appropriately prepared to create an individual career~~
20 ~~and academic plan pursuant to [section 279.61](#), incorporate~~
21 ~~foundational career and technical education concepts aligned~~
22 ~~with the six career and technical education service areas as~~
23 ~~defined in [subsection 5](#), paragraph "h", incorporate relevant~~
24 ~~twenty-first century skills to facilitate career readiness, and~~
25 ~~introduce students to career opportunities within the local~~
26 ~~community and across this state.~~

27 Sec. 7. Section 256.11, subsection 4, Code 2026, is amended
28 by adding the following new paragraph:

29 NEW PARAGRAPH. 0c. Career exploration and development
30 shall be designed so that students are appropriately prepared
31 to create an individual career and academic plan pursuant to
32 [section 279.61](#), incorporate foundational career and technical
33 education concepts aligned with the six career and technical
34 education service areas as defined in subsection 5, paragraph
35 "h", incorporate relevant twenty-first century skills to

1 facilitate career readiness, and introduce students to career
2 opportunities within the local community and across this state.

3 Sec. 8. Section 256.11, subsection 5, paragraph a, Code
4 2026, is amended by adding the following new subparagraph:

5 NEW SUBPARAGRAPH. (3) The five units of science may include
6 not more than one unit of instruction related to computer
7 science and artificial intelligence.

8 Sec. 9. Section 256.11, subsection 5, paragraphs e and l,
9 Code 2026, are amended to read as follows:

10 e. Two additional units of mathematics.

11 (1) The two additional units of mathematics may include
12 instruction related to applied sciences, technology,
13 engineering, or manufacturing, including transportation,
14 distribution, logistics, architecture, and construction, which
15 may be offered and taught through dedicated units of coursework
16 or through units of coursework that also meet the requirements
17 of the coursework required under paragraph "h", subparagraph
18 (1), subparagraph division (c). ~~The state board shall adopt~~
19 rules pursuant to ~~chapter 17A~~ to implement this paragraph.

20 (2) The two additional units of mathematics may include not
21 more than one unit of instruction related to computer science
22 and artificial intelligence.

23 (3) The state board shall adopt rules pursuant to chapter
24 17A to implement this paragraph.

25 l. (1) One-half unit of computer science commencing
26 with the school year beginning July 1, 2022 and artificial
27 intelligence. The one-half unit of computer science and
28 artificial intelligence shall incorporate the standards
29 established pursuant to ~~section 256.7, subsection 26~~, paragraph
30 "a", subparagraph (4), and may be offered online in accordance
31 with rules adopted pursuant to ~~section 256.7, subsection 32~~,
32 paragraph "a". The one-half unit of computer science and
33 artificial intelligence shall satisfy all of the following
34 requirements:

35 (a) Be of high quality.

1 (b) Meet or exceed the curriculum standards and
2 requirements established by the state board.

3 (c) Be offered and taught in a traditional classroom
4 setting; provided, however, that if a traditional classroom
5 setting is not feasible, the one-half unit of computer science
6 and artificial intelligence may be offered and taught in any
7 of the following ways:

8 (i) Through a combination of instruction that is delivered
9 in a traditional classroom setting and instruction that is
10 delivered primarily over the internet.

11 (ii) Through instruction that is delivered primarily over
12 the internet in accordance with rules adopted by the state
13 board pursuant to section 256.7, subsection 32, paragraph
14 "a", and is tailored to meet the needs of each participating
15 student.

16 (2) (a) All students, beginning with students in the
17 2030-2031 school year graduating class, shall complete one-half
18 unit of computer science and artificial intelligence as a
19 condition of graduation, which shall incorporate the standards
20 established pursuant to section 256.7, subsection 26, paragraph
21 "a", subparagraph (4), and which may include computer science
22 instruction provided in grade eight pursuant to section
23 256.11, subsection 4. The condition of graduation shall not
24 apply to a student who enrolls in a school district or an
25 accredited nonpublic school in grade twelve immediately after
26 being enrolled in a school located outside of this state that
27 does not impose a comparable computer science and artificial
28 intelligence instruction requirement.

29 (b) (i) On or before July 1, 2030, a school district or
30 accredited nonpublic school may apply to the department of
31 education for a one-time waiver of the implementation of the
32 condition of graduation described in this subparagraph. The
33 waiver shall be for a period not longer than one school year.

34 (ii) The state board shall adopt rules pursuant to chapter
35 17A establishing standards for the submission of applications

1 for waivers of the implementation of the condition of
2 graduation described in this subparagraph and the department's
3 consideration and approval of waivers. The rules must
4 require that the department of education may only approve an
5 application for a waiver submitted pursuant to subparagraph
6 subdivision (i) if the school district or accredited nonpublic
7 school applying for the waiver demonstrates to the department
8 that the school district or accredited nonpublic school made
9 a reasonable attempt to implement the condition of graduation
10 described in this subparagraph.

11 (iii) This subparagraph division is repealed July 1, 2031.

12 (3) Each school district shall report annually to the
13 department all of the following:

14 (a) The course names and course codes associated with all
15 computer science and artificial intelligence courses offered
16 and taught in the school district pursuant to this paragraph
17 and, to the extent such information is available, a description
18 of the format in which each course is offered and taught.

19 (b) The number of teachers employed by the school district
20 who teach computer science and artificial intelligence
21 instruction, sortable by the endorsements and authorizations
22 held by each teacher and the highest academic degree attained
23 by each teacher. The report shall not include any personally
24 identifiable information related to the teachers employed by
25 the school district who teach computer science and artificial
26 intelligence instruction.

27 (c) The number of students enrolled in a computer science
28 and artificial intelligence course offered and taught by the
29 school district pursuant to this paragraph, sortable by the
30 grade level of each student. The report shall not include any
31 personally identifiable information related to the students
32 enrolled in a computer science and artificial intelligence
33 course offered and taught by the school district.

34 Sec. 10. Section 256.11, subsection 5, paragraph h, Code
35 2026, is amended by adding the following new subparagraph:

1 NEW SUBPARAGRAPH. (04) Units of instruction provided
2 under subparagraph (1), subparagraph division (b) or (c), may
3 include up to one unit of instruction related to computer
4 science and artificial intelligence, if the teacher providing
5 such instruction has either an initial career and technical
6 education authorization issued by the board of educational
7 examiners or a career and technical education authorization
8 issued by the board of educational examiners.

9 Sec. 11. Section 256.11, subsection 17, paragraph a,
10 subparagraph (2), Code 2026, is amended to read as follows:

11 (2) If a school district or accredited nonpublic school
12 meets the requirements of subparagraph (1), subparagraph
13 division (a) or (b), the school district or accredited
14 nonpublic school may exceed the two-subject limitation
15 specified in subparagraph (1), unnumbered paragraph 1, for the
16 purpose of providing world language, personal finance literacy,
17 and computer science and artificial intelligence coursework
18 online in accordance with paragraph "c".

19 Sec. 12. Section 256.13, Code 2026, is amended to read as
20 follows:

21 **256.13 Nonresident pupils.**

22 The boards of directors of two or more school districts may
23 by agreement provide for attendance of pupils residing in one
24 district in the schools of another district for the purpose of
25 taking courses not offered in the district of their residence,
26 including computer science and artificial intelligence courses
27 described in section 256.11, subsection 5, paragraph "1". The
28 boards may also provide by agreement that the districts will
29 combine their enrollments for one or more grades. Courses
30 and grades made available to students in this manner shall be
31 considered as complying with any standards or laws requiring
32 the offering of such courses and grades. The boards of
33 directors of districts entering into such agreements may
34 provide for sharing the costs and expenses of the courses. If
35 the agreement provides for whole grade sharing, the costs and

1 expenses shall be paid as provided in sections 282.10 through
2 282.12.

3 Sec. 13. Section 256.16, subsection 1, Code 2026, is amended
4 by adding the following new paragraph:

5 NEW PARAGRAPH. *p.* Include preparation in computer science,
6 artificial intelligence, and computational thinking that is
7 appropriate to the grade level and subject area in which
8 the student who is enrolled in the practitioner preparation
9 program will teach. A higher education institution providing
10 practitioner preparation may satisfy this requirement by
11 requiring a student who is enrolled in the practitioner
12 preparation program to complete a dedicated course or
13 module, or by integrating such preparation into the existing
14 practitioner preparation program.

15 Sec. 14. Section 256.146, subsection 19, Code 2026, is
16 amended to read as follows:

17 19. Adopt rules pursuant to [chapter 17A](#) establishing
18 endorsements and authorizations for computer science and
19 artificial intelligence instruction, including. The rules
20 shall include traditional and nontraditional pathways for
21 obtaining such endorsements or authorizations, including
22 through instruction that is delivered primarily over the
23 internet that may be used to demonstrate competence in computer
24 science and artificial intelligence instruction for teachers
25 under an initial or conditional license.

26 Sec. 15. Section 262.9, Code 2026, is amended by adding the
27 following new subsection:

28 NEW SUBSECTION. 43. Develop, in consultation with the
29 institutions of higher education governed by the board,
30 guidance related to how a higher education institution
31 providing practitioner preparation can best incorporate
32 preparation related to computer science, artificial
33 intelligence, and computational thinking into the approved
34 practitioner preparation program, as required under section
35 256.16, subsection 1, paragraph "p", and distribute such

1 guidance to each higher education institution providing
2 practitioner preparation.

3 Sec. 16. Section 280.3, subsection 3, Code 2026, is amended
4 to read as follows:

5 3. The board of directors of each public school district
6 and the authorities in charge of each nonpublic school shall
7 develop and implement a kindergarten through grade twelve
8 computer science and artificial intelligence plan ~~by July 1,~~
9 ~~2022, which~~ that incorporates the standards established under
10 section 256.7, subsection 26, paragraph "a", subparagraph (4),
11 and the minimum educational standards relating to computer
12 science and artificial intelligence contained in section
13 256.11.

14 Sec. 17. Section 284.6A, subsection 1, paragraphs a and b,
15 Code 2026, are amended to read as follows:

16 a. That instruction in high-quality computer science and
17 artificial intelligence for elementary, middle school, and
18 high school students establishes a foundation for personal and
19 professional success in a high-technology, knowledge-based Iowa
20 economy.

21 b. It is the goal of the general assembly that ~~by July~~
22 ~~1, 2019,~~ each accredited high school offer at least one
23 high-quality computer science and artificial intelligence
24 course, each accredited middle school offer instruction in
25 exploratory computer science and artificial intelligence, and
26 each accredited elementary school offer instruction in the
27 basics of computer science and artificial intelligence.

28 Sec. 18. Section 284.6A, subsection 1, Code 2026, is amended
29 by adding the following new paragraph:

30 NEW PARAGRAPH. *0c.* In order to provide students with the
31 competencies to compete in a twenty-first century workforce,
32 develop skills that translate into high-wage jobs, develop
33 problem solving and critical thinking skills, and encourage
34 the creation and retention of new, high-wage jobs in this
35 state, all high school students enrolled in school districts

1 or accredited nonpublic schools, beginning with the 2030-2031
2 school year graduating class, students should complete one
3 semester of computer science and artificial intelligence
4 as a condition of graduation as required by section 256.7,
5 subsection 26, paragraph "a".

6 Sec. 19. Section 284.6A, subsection 2, paragraphs a and b,
7 Code 2026, are amended to read as follows:

8 a. A school district or accredited nonpublic school, or
9 a collaborative of one or more school districts, accredited
10 nonpublic schools, and area education agencies, may apply to
11 the department, in the manner prescribed by the department, to
12 receive moneys from the fund to provide proven professional
13 development activities for Iowa teachers in the area of
14 computer science and artificial intelligence education.

15 b. A school district or accredited nonpublic school may
16 apply to the department, in the manner prescribed by the
17 department, to receive moneys from the fund to provide tuition
18 reimbursement for Iowa teachers, including Iowa teachers under
19 an initial or conditional license, seeking endorsements or
20 authorizations for computer science and artificial intelligence
21 under [section 256.146, subsection 19](#).

22 Sec. 20. STATE MANDATE FUNDING SPECIFIED. In accordance
23 with section 25B.2, subsection 3, the state cost of requiring
24 compliance with any state mandate included in this Act shall
25 be paid by a school district from state school foundation aid
26 received by the school district under section 257.16, and from
27 any grants received under the computer science professional
28 development fund established pursuant to section 284.6A. This
29 specification of the payment of the state cost shall be deemed
30 to meet all of the state funding-related requirements of
31 section 25B.2, subsection 3, and no additional state funding
32 shall be necessary for the full implementation of this Act
33 by and enforcement of this Act against all affected school
34 districts.

35 Sec. 21. APPLICABILITY. The following applies to students

1 admitted to an approved practitioner preparation program on or
2 after July 1, 2028:

3 The section of this Act amending section 256.16.

4 EXPLANATION

5 The inclusion of this explanation does not constitute agreement with
6 the explanation's substance by the members of the general assembly.

7 This bill relates to computer science and artificial
8 intelligence education, including high school curricula and
9 graduation requirements and practitioner preparation programs.

10 Current law provides that the graduation requirements
11 applicable to students enrolled in school districts and
12 accredited nonpublic schools shall include, at a minimum,
13 satisfactory completion of four years of English and language
14 arts, three years of mathematics, three years of science,
15 and three years of social studies. The bill provides that
16 such graduation requirements shall also include, beginning
17 with students in the 2030-2031 school year graduating class,
18 satisfactory completion of one semester of computer science
19 and artificial intelligence, or the trimester or quarter
20 equivalent, which may include computer science and artificial
21 intelligence instruction provided in grade eight. The bill
22 also provides that one semester, or the trimester or quarter
23 equivalent, of computer science and artificial intelligence
24 instruction satisfies the graduation requirement associated
25 with both computer science and artificial intelligence
26 instruction and one semester, or the trimester or quarter
27 equivalent, of the graduation requirement associated with
28 either mathematics or science. Current law also requires that
29 the graduation requirements provide for the establishment
30 of high-quality standards for computer science education
31 taught by elementary, middle, and high schools. The bill
32 provides that these standards must require that all levels
33 and grades of computer science and artificial intelligence
34 instruction include instruction related to the fundamental
35 concepts of evolving computer science technologies and

1 artificial intelligence, including instruction that explains
2 what artificial intelligence is, how artificial intelligence
3 works, and the societal and ethical impacts of artificial
4 intelligence.

5 The bill provides that the condition of graduation shall
6 not apply to a student who enrolls in a school district or
7 an accredited nonpublic school in grade 12 immediately after
8 being enrolled in a school located outside of this state that
9 does not impose a comparable computer science and artificial
10 intelligence instruction requirement. The bill authorizes
11 school districts and accredited nonpublic schools to apply
12 to DE for a one-time waiver of the implementation of the
13 graduation requirement. The bill requires the state board
14 of education to adopt rules establishing standards for the
15 submission of applications for waivers of the implementation of
16 the condition of graduation and the department of education's
17 (DE) consideration and approval of waivers.

18 The bill requires the director of DE to publish on DE's
19 internet site, on or before June 30, 2027, a list of computer
20 science and artificial intelligence courses that satisfy the
21 rules adopted by the state board and that a school district or
22 accredited nonpublic school may use to implement the graduation
23 requirement. Additionally, the director is required to create
24 and publish on DE's internet site, on or before June 30, 2027,
25 a plan to expand statewide computer science and artificial
26 intelligence teacher capacity to support implementation of the
27 graduation requirement. The plan must specifically address
28 targeted support for schools with a total enrollment of less
29 than 500 students in grades 9 through 12.

30 Current law requires school districts and accredited
31 nonpublic schools to offer and teach to students enrolled in
32 grades 9 through 12 five units of science and two additional
33 units of mathematics. The bill provides that the five units
34 of science may include not more than one unit of instruction
35 related to computer science and artificial intelligence, and

1 the two additional units of mathematics may include not more
2 than one unit of instruction related to computer science and
3 artificial intelligence.

4 Current law requires school districts and accredited
5 nonpublic schools to offer and teach to students enrolled in
6 grades 9 through 12 one-half unit of computer science. The
7 bill modifies provisions related to how the computer science
8 and artificial intelligence instruction is offered and taught
9 by allowing the instruction to be offered and taught through
10 blended learning, or through instruction that is delivered
11 primarily over the internet, when it is not possible to provide
12 such instruction in a traditional classroom setting. Current
13 law allows school districts to, by agreement, provide for
14 attendance of pupils residing in one district in the schools
15 of another district for the purpose of taking courses not
16 offered in the district of their residence. The bill provides
17 that such courses may include computer science and artificial
18 intelligence courses.

19 The bill modifies provisions related to the career and
20 technical education instruction provided to students enrolled
21 in grades 9 through 12 to provide that instruction provided
22 under either the arts, communications, and information systems
23 or applied sciences, technology, engineering, and manufacturing
24 service areas may include up to one unit of instruction
25 related to computer science and artificial intelligence, if the
26 teacher has either an initial career and technical education
27 authorization or a career and technical education authorization
28 issued by the BOEE.

29 The bill requires school districts to report annually to
30 DE specified information related to computer science and
31 artificial intelligence courses offered and taught in the
32 school district, teachers providing computer science and
33 artificial intelligence instruction, and students involved
34 in computer science and artificial intelligence courses.
35 DE is required to publish on DE's internet site all of the

1 information submitted by school districts pursuant to these
2 provisions.

3 The bill requires each higher education institution
4 providing practitioner preparation to include preparation in
5 computer science, artificial intelligence, and computational
6 thinking that is appropriate to the grade level and subject
7 area in which the student who is enrolled in the practitioner
8 preparation program will teach. This requirement applies to
9 students admitted to an approved practitioner preparation
10 program on or after July 1, 2028. The bill requires the
11 state board of regents, in consultation with the institutions
12 of higher education governed by the state board of regents,
13 to develop guidance related to how a higher education
14 institution providing practitioner preparation can best
15 incorporate preparation related to computer science, artificial
16 intelligence, and computational thinking into the approved
17 practitioner preparation program, as required under the bill's
18 provisions.

19 The bill requires that the rules adopted by the BOEE
20 establishing nontraditional pathways for obtaining endorsements
21 and authorizations for computer science and artificial
22 intelligence instruction also include opportunities to obtain
23 such endorsements or authorizations through instruction that
24 is delivered primarily over the internet that may be used to
25 demonstrate competence in computer science and artificial
26 intelligence instruction for teachers under an initial or
27 conditional license.

28 Current law establishes a computer science professional
29 development incentive fund in the state treasury under the
30 control of DE. Current law authorizes DE to disburse moneys
31 contained in the fund to a school district or accredited
32 nonpublic school to provide tuition reimbursement for Iowa
33 teachers seeking endorsements or authorizations for computer
34 science and artificial intelligence from the BOEE. The bill
35 modifies this provision to allow such tuition reimbursement to

1 be provided to Iowa teachers under an initial or conditional
2 license.

3 The bill may include a state mandate as defined in Code
4 section 25B.3. The bill requires that the state cost of
5 any state mandate included in the bill be paid by a school
6 district from state school foundation aid received by the
7 school district under Code section 257.16, and from any grants
8 received under the computer science professional development
9 fund. The specification is deemed to constitute state
10 compliance with any state mandate funding-related requirements
11 of Code section 25B.2. The inclusion of this specification is
12 intended to reinstate the requirement of political subdivisions
13 to comply with any state mandates included in the bill.
14 The bill makes conforming changes.